What Single-Channel and Low-Cost EEG Can and Cannot Measure: A Reproducible Degradation Benchmark and Reporting Protocol

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Abstract

Low-cost EEG headsets are used to make claims about attention, relaxation, sleep and emotion, usually without the information needed to judge whether a recording can support them. We present a reproducible benchmark that degrades open multi-channel recordings to single-channel consumer conditions — one electrode, 128 Hz sampling, 1 µV quantisation, 1–40 Hz band-limiting, pink sensor noise at 20/10/5/0 dB, synthetic blinks at 5/15/30 per minute and a combined condition — and measures which standard analyses survive. Using all 109 subjects of the EEG Motor Movement/Imagery Dataset, alpha reactivity was essentially unaffected by the acquisition chain once the reference was taken at the same electrode (99.99% retained; 95% CI of the retention ratio $[0.9996, 1.0001]$; the outcome is a ratio of ratios, so gain-like steps cancel by construction), whereas the electrode-site choice cost about 42% of eyes-closed alpha power (Fz/posterior ratio 0.58). Noise reduced the effect monotonically, to about 0.59 of the reference at 0 dB (0.68 of the clean chain); a simple additive blink model changed it by about 0.2%, which reflects the construction of the model rather than a finding about real blinks. Using 18 high-end BrainVision recordings from an open affective dataset, valence, arousal and four-quadrant decoding from the full montage reached 0.69, 0.68 and 0.58 (chance 0.50 and 0.25). Reducing the same data to one electrode did not detect the valence effect from one electrode (0.52; no significant evidence above chance, equivalence not tested), reduced arousal to 0.56 (q=0.050 in the six-contrast family; not significant in the site sweep) and four-class accuracy to 0.34 (above chance but far below the montage); valence was not distinguishable from chance at any single site, and four-class accuracy stayed at 0.33–0.38 across Fz, Cz, Pz and Oz. A band control showed that the full-montage advantage is partly carried by beta-band features (theta+alpha only: 0.60 valence, 0.46 four-class; beta only: 0.70 and 0.62), a pattern consistent with a possible muscular contribution rather than proof of a cortical mechanism. A low-cost 14-channel headset performed at a similar level to the matched high-end montage (0.67/0.71 versus 0.73/0.66 on the 12 shared participants), although a ±10-point equivalence test did not pass. Short-lag alpha stability was low ($r_{SB} = 0.016$) while block-level alpha power was stable (ICC(2,1)= 0.92), and the motor arm was invalid because its reference decoder did not exceed chance. The deliverable is a decision matrix, a single-channel reporting checklist, and a seeded analysis pipeline with the JSON files behind every number; a public archive DOI will be deposited before publication.

Keywords: consumer EEG, single-channel, reproducibility, benchmark, affective decoding, reporting standards

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1 Introduction

Consumer EEG is everywhere; its evidence base is not

Low-cost EEG headsets are now sold as attention trainers, relaxation aids, sleep gadgets and emotion sensors. The hardware is real: consumer devices can record above the noise floor of the relevant bands, and some have been validated against research amplifiers for event-related potentials [18, 1], while resting-state comparisons report differences in spectral estimates and in test–retest behaviour across systems [28, 5]. What is usually missing is the information needed to decide whether a given recording supports a given claim: which electrode, which sampling rate, which filters, how artifacts were handled, how many trials, and how the feature relates to a full-montage reference.

This gap has two consequences. First, results are hard to reproduce: a claim such as "alpha power increases during relaxation" can be produced by a well-configured research system or by a badly filtered consumer recording, and the paper often does not let a reader tell the difference. Second, the same gap makes it easy to overclaim: a single frontal channel measuring a mixture of cortical activity, eye movements and muscle noise [37] is presented as if it measured a specific cognitive process.

Why a benchmark rather than another device comparison

Device-comparison studies answer "is headset X close to amplifier Y?" under one laboratory protocol. They do not answer the question a reader of a consumer-EEG paper actually has: *given this acquisition setup, which of the standard analyses still work?* Answering that requires systematically degrading high-quality recordings along the dimensions consumer hardware changes — channel count, sampling rate, amplitude resolution, bandwidth, noise, ocular contamination — and measuring which analyses survive, with a full-montage analysis as the reference ceiling.

Open datasets make this possible without new participants. The EEG Motor Movement/Imagery Dataset provides 109 subjects, 64 channels and both eyes-open and eyes-closed baselines [34]; an open low-cost-versus-high-end affective dataset provides labelled emotional trials recorded with a high-end system [32, 31]. Both are widely used, publicly documented and have known acquisition parameters.

Contributions

1. A **reproducible degradation pipeline** that converts multi-channel recordings into single-channel consumer conditions in explicit, separately reported steps (channel selection, 128 Hz resampling, 1 µV quantisation, 1–40 Hz band-limiting, pink sensor noise at four SNR levels, synthetic blink contamination at three rates, and a combined condition).

2. A **planned benchmark** of four analysis families: alpha reactivity, affective decoding, motor decoding (exploratory) and reliability.

3. Two results that point in opposite directions. Alpha reactivity is essentially unaffected by the acquisition chain once the reference is site-matched (99.99% retained), so the chain itself is not the limiting factor; electrode choice and noise are. Affective valence decoding is above chance from a full montage (0.69) but is not detected from a single electrode (0.52); arousal is reduced (0.68 to 0.56) rather than abolished.

4. An **open reporting checklist** for single-channel studies, derived from COBIDAS MEEG [24], EEG-BIDS [25] and, as a general reporting model, the CRED-nf checklist [30], and released with code and derived data.

We emphasise what this paper is *not*: it is not a clinical or wellness intervention study, it does not validate any commercial product, and it makes no claim that a particular headset is "good" or "bad". It provides a reference against which such claims can be evaluated.

2 Related Work

Consumer and low-cost EEG validation

Comparisons of medical-grade and consumer wireless systems report that consumer hardware can recover common event-related components under favourable conditions, while signal quality and artifact susceptibility differ substantially between devices and sessions [28, 18, 33]. One of the closest prior studies compared four systems (including MindWave and Muse) on five participants across two visits, focusing on Fp1 and explicitly comparing eyes-closed and eyes-open spectra and their test–retest reliability [28]. That is close to our alpha arm on real hardware, so the gap cannot be "nobody has looked at consumer spectral reliability"; the gap is that existing work does not quantify which analyses survive a *controlled* degradation chain, and does not connect that chain to claim-level verdicts. Work on channel subsets for emotion decoding has shown that small, well-chosen montages can be competitive with large ones [38]; what is missing is a controlled degradation of the same recordings with a site-matched reference and a claim-level decision rule. Our contribution is therefore the decomposition and the decision matrix, not the observation that fewer channels can hurt.

Reproducibility and reporting standards

Psychology and neuroscience face a well-documented reproducibility problem [23]. In electrophysiology the community response has been standardised reporting (COBIDAS MEEG) [24] and standardised data organisation (EEG-BIDS) [25]. Both are written primarily for research-grade acquisitions; they specify what to report, but not what can be inferred when the acquisition is a single channel. The present benchmark is designed to fill that inference gap and to make the reporting checklist operational for consumer-device studies.

Analysis pipelines and artifact handling

Modern open pipelines make preprocessing explicit and reproducible [10, 8, 22], and automated artifact rejection and independent-component classification reduce researcher degrees of freedom [14, 27]. These tools assume enough channels for spatial decomposition; with one channel, ICA is unavailable and most artifact rejection reduces to amplitude thresholds or regression against auxiliary signals. A benchmark of single-channel conditions therefore must treat artifact handling as part of the independent variable, not as a preprocessing detail.

Feature validity and open data

Band-power features are the workhorse of consumer-EEG claims, and their test–retest reliability is a prerequisite for any individual-level interpretation. Open datasets with documented conditions and affective labels allow reliability and decoding to be measured rather than asserted [34, 17]; the labels are normative picture categories, not participants' own reports. We build directly on the EEGBCI resources and on an open affective dataset that contains both low-cost and high-end recordings [32, 31]; DEAP and DREAMER are discussed as alternatives that could not be used. We report every degradation step so that a reader can map our results onto their own acquisition.

The affective dataset used here has its own decoding study [32], which compared the low-cost and high-end systems under three channel-count and sampling-rate scenarios and reported F1 values for four-class classification. Our analysis is not a replication of that study: we use

balanced accuracy, within-subject cross-validation, a fully crossed degradation chain that includes quantisation, noise and blink arms, a same-site reference, and claim-level decision rules. The F1 values in the source study and the balanced accuracies here are therefore not directly comparable, and we do not interpret our lower values as contradicting it. What this paper adds is the decomposition of where performance goes when the representation is reduced, not a new headline accuracy for the dataset.

Positioning

Relative to device-comparison work, our contribution is analysis-centred and degradation-based. Relative to reporting standards, it is quantitative: rather than saying which variables to report, it estimates what each variable is worth under single-channel constraints. To our knowledge, no existing benchmark reports, for the same open datasets, how alpha reactivity, single-trial decoding and reliability change as a function of explicit consumer-hardware degradations, and which of those changes survive a claim-level decision rule. Blink susceptibility is a stated limitation of this version and is not claimed as a measured outcome.

Benchmarking methodology

The idea of comparing analysis pipelines on shared datasets is established in BCI research: MOABB provides trustworthy benchmarking across datasets and pipelines [15]. Our benchmark differs in object — it holds the pipeline approximately fixed and varies the acquisition conditions, rather than comparing algorithms — but it inherits the principle that a claim about a method should be evaluated against a common, open reference. Filter design is a concrete example of why this matters: apparently small high-pass choices can create artifactual effects and incorrect conclusions [36], which is exactly the kind of degradation our chain isolates and reports.

3 Methods

Datasets

EEG Motor Movement/Imagery Dataset [34]. All 109 subjects were used: 64 channels at 160 Hz, runs 1–2 for the eyes-open and eyes-closed baselines and runs 4, 8 and 12 for imagined left/right hand movement, as defined in the dataset documentation. Motor decoding uses the ten subjects for whom the imagery runs were degraded in this pass, and is treated as exploratory.

Affective dataset [32, 31]. The open CC BY 4.0 dataset from Zenodo 14787743 contains EEG from 20 participants who viewed International Affective Picture System images selected to induce four valence–arousal quadrants, recorded with a low-cost Emotiv EPOC+ and a high-end BrainVision system. The archive's `EEG/Experiment_files` directory contains 18 BrainVision recordings (66 channels, 500 Hz, sampling interval 2000 µs); a nineteenth recording sits in a separate `EEG registro_nuevo` folder and was not used. All 18 recordings in the analysed directory completed the pipeline. The files are named `eegbci<subject>-<session>` by the archive; this is the archive's own naming for the affective recordings and does not refer to the EEGBCI motor dataset. The valence/arousal labels come from the affective picture protocol, not from a motor task. BrainVision marker codes map to the quadrants as follows: 1 = high valence/high arousal, 2 = low valence/high arousal, 3 = high valence/low arousal, 4 = low valence/low arousal, 10/14 = neutral. Each recording contains approximately 10 trials per quadrant and 8 neutral trials; neutral trials are excluded from valence and arousal analyses. Quadrant labels are the normative assignments of the International Affective Picture System [20], not participants' own ratings. Marker positions are used as onsets and each trial is epoched 0–2.5 s after onset. The low-cost Emotiv arm is analysed for the 12 participants whose CSV files are usable in the distributed

archive (five files are empty or header-only and three fail the timestamp alignment). The device has been used in validation work on event-related potentials [1]. The files contain 14 EEG channels at 128 Hz in device units with absolute wall-clock timestamps. The log session whose absolute trial times fall inside the CSV span is selected; trials are aligned by converting each log onset to an absolute time (SessionStartDateTimeUtc plus the millisecond onset) and matching it to the nearest CSV sample, with a clock offset chosen from $\{0, \pm 1, \pm 2\}$ hours by maximising the number of trials inside the recording span. Emotiv epochs run from -0.2 to $+2.3$ s around the aligned sample (the BrainVision epochs run 0–2.5 s), and the same band-power and LDA pipeline is applied. The median absolute difference between a log onset and its nearest CSV sample was 25 ms (95th percentile 59 ms, maximum 70 ms; stored per subject in `results/emotiv.json`), and no trial exceeded the 1 s alignment screen (this is a timestamp screen, not an amplitude rejection threshold). The two condition arms in the archive are not otherwise distinguished. This yields a three-way comparison: high-end full montage, low-cost 14-channel headset, and high-end data reduced to a single degraded channel.

Both datasets are de-identified and openly available. No new data were collected.

Degradation grid

EEGBCI recordings are processed at a full-montage reference and at twelve single-channel conditions: a cumulative acquisition chain (Fz; +128 Hz; +1 µV quantisation; +1–40 Hz band-limiting), a noise arm (20/10/5/0 dB pink noise, no blinks; pink noise is a conservative broadband proxy and real consumer noise also contains white electronics noise, mains interference and drift, so the noise arm should be read as one plausible contamination profile rather than a device model), a blink arm (5/15/30 per minute, no noise) and a combined arm (0 dB + 30 blinks/min). The single electrode is selected by canonical name matching because EEGBCI channel labels carry punctuation (e.g. `Fz.`); the chosen label is written to the condition metadata. Blink transients are 350 ms Hann-windowed 2 Hz templates scaled to four standard deviations of the channel; with one channel the template cannot be given a frontal topography, which is a limitation of the model.

For the affective dataset the same single-channel chain (128 Hz, 1 µV, 1–40 Hz, Fz) is applied to the high-end recording and compared with the full montage. The noise and blink arms are not applied to the affective data; that comparison is left to future work.

Site-matched sensitivity analysis

A single Fz electrode and a 64-channel average are not the same measurement: alpha is posterior-dominant, so comparing them confounds channel count with electrode site. We therefore compute, in the eyes-closed run, Fz power from the full-montage recording, Fz power after the full single-channel chain, the 64-channel average and the average over posterior channels. The site-matched ratio (single-channel Fz versus full-montage Fz) is the primary estimate of what the acquisition chain costs; the scalp-average comparison is reported as the naive estimate that a reader would get without site-matching.

Analyses

A. Alpha reactivity. Band power (8–12 Hz) is computed with Welch's method (2 s windows, 50% overlap) on continuous data; the outcome is the natural-log ratio of eyes-closed to eyes-open power, the classical resting-state contrast [2]. The primary retention estimator is the geometric mean ratio derived from the paired subject-level log difference. For the acquisition chain the reference is the same-site full recording (Fz in the full montage); for the noise, blink and combined arms the reference is the clean single-channel chain, because those manipulations are incremental contamination of that chain. A degradation claim is reported as *supported* when the *lower* 95%

confidence bound of the relevant retention ratio exceeds 0.50; as *not supported* when the lower bound is at or below 0.50; and the decision matrix reports the retention interval alongside the verdict. The 50% threshold is a planned convention of this benchmark, not a literature-derived constant, and it is not a formal non-inferiority test. The pre-specified 60% practical floor for balanced accuracy is reported separately as a stricter auxiliary criterion. Absolute powers are released so readers can evaluate alternative summaries.

B. Motor decoding (exploratory). Single-trial log band power (8–30 Hz; mu/beta event-related desynchronisation [26]) with shrinkage LDA and 5-fold stratified within-subject cross-validation. A CSP-based reference decoder (CSP fitted inside each fold) is reported as a sensitivity analysis. Because CSP requires multiple channels, the single-channel condition cannot receive the same spatial filter; that asymmetry is part of the constraint quantified here. The arm is reported as invalid unless a reference decoder's 95% CI excludes chance (0.50). Because nominal chance levels can be exceeded by chance, interpretation uses empirical confidence intervals rather than the theoretical level [6, 4].

C. Affective decoding. For each recording, log band power in theta (4–8 Hz), alpha (8–12 Hz) and beta (13–30 Hz) is extracted per channel and trial; the classifier is shrinkage LDA with 5-fold stratified within-subject cross-validation. Three targets are decoded: valence (quadrants 1/3 versus 2/4), arousal (1/2 versus 3/4) and the four quadrants. The same pipeline is run (i) on the full montage, (ii) after each cumulative acquisition step (128 Hz, quantisation, band-limiting), (iii) on single sites Fz, Cz, Pz and Oz, and (iv) on the full montage with only theta+alpha features and with only beta features. The last two conditions separate a representation effect from a band/EMG effect. Chance is 0.50 for the binary targets and 0.25 for the four-class target; the empirical CI is the inferential criterion [6]. The low-cost Emotiv arm is analysed separately and compared with the high-end arm on the 12 participants present in both, using a TOST procedure for equivalence.

D. Reliability. Alpha power is computed in consecutive 2 s windows of the eyes-closed run; odd and even windows are correlated, and the Spearman–Brown formula is reported as a descriptive *short-lag stability* measure. Odd and even windows are about two seconds apart, so this is not a split-half reliability coefficient and is not directly comparable to the ICC. In the ten-subject subset with three repeated imagined-hand runs, ICC(2,1) [35] of alpha power across runs is reported as a block-level estimate. Neither is a test–retest coefficient across sessions, which is not identifiable in these datasets.

E. Blink susceptibility of band ratios. Pre-specified but not implemented; reported as untested.

Statistical analysis

Subject-level outcomes are summarised with means and 95% confidence intervals (t-distribution, $df = n - 1$). Condition effects are reported both as unpaired means and as paired contrasts against the reference for the same subjects. Benjamini–Hochberg FDR [3] is applied within the alpha condition family and within the affective family; both raw and adjusted values are released. Mixed-effects models are not used here, and the exploratory nature of the study is stated explicitly. Equivalence between the low-cost and high-end devices is tested with a two one-sided tests (TOST) procedure using a ± 0.10 balanced-accuracy margin [19].

Reproducibility

All degradation parameters, random seeds, dataset file hashes (written by `scripts/01_download.py`), analysis scripts and derived feature tables are included in the repository. Raw datasets are not redistributed; download scripts reproduce the exact files used. Figure source data are archived with the results JSON. Artifact handling is deliberately minimal: no Autoreject or ICA is applied, because single-channel data cannot support ICA and the benchmark is about the ac-

quisition conditions rather than about cleaning; the ocular-artifact literature [11, 7] and the EMG-contamination literature [37, 9, 21] are discussed as limitations. Single-channel test–retest work reports eyes-closed relative-power ICCs of roughly 0.76–0.85 for a consumer device [29]; our short-lag estimate is a within-session stability measure and is not directly comparable.

4 Results

Channel reduction, bandwidth and quantisation do not degrade alpha reactivity

At the full montage the eyes-closed/eyes-open alpha log ratio was 0.91 (95% CI $[0.78, 1.04]$; $N = 109$). The Fz single-channel chain — single electrode, 128 Hz, 1 µV quantisation, 1–40 Hz band-limiting — gave 0.77 ($[0.65, 0.89]$). Comparing the single channel with the scalp average, the paired difference was -0.14 ($[-0.19, -0.09]$). The mean per-subject retention ratio was 0.90 ($[0.85, 0.96]$) and the geometric estimator (ratio of means) was 0.87; the two estimators differ because ratios are skewed and both are reported in `results/paired.json` and `inference.json`.

That comparison is confounded by electrode site. Two corrections are needed. First, the acquisition chain itself costs almost nothing to *reactivity*: keeping the same electrode (Fz in the full recording versus Fz after the chain), the paired difference was -0.0001 ($[-0.0004, 0.0001]$), i.e. **99.99%** of the reactivity effect was retained; the corresponding retention interval $[0.9996, 1.0001]$ is stored in `results/site_metrics.json` (`chain_retention`). Second, the electrode-site decision is not free: eyes-closed alpha power at Fz is only about **58%** of the posterior-channel average (ratio 0.58, $[0.52, 0.63]$). A single-channel device therefore pays for its site choice, not for its sampling or bandwidth. This also explains why 128 Hz sampling, 1 µV quantisation and a 40 Hz low-pass left the estimate essentially unchanged: the band versus quantisation contrast was -0.00012 (retention 0.9999, $p = 0.007$), which is negligible for 8–12 Hz Welch power and consistent with earlier filter work [36].

Table 1 reports the condition means and Figure 1 shows the degradation grid, while Figure 2 shows retention.

Table 1: Alpha reactivity by condition ($N = 109$; mean log ratio with 95% CI). Noise and blink arms are separate; the final row combines both.

Condition	Log ratio	95% CI
Full montage (64 ch average)	0.91	[0.78, 1.04]
Single channel (Fz) chain	0.77	[0.65, 0.89]
+ noise 20 dB	0.76	[0.64, 0.88]
+ noise 10 dB	0.70	[0.58, 0.83]
+ noise 5 dB	0.60	[0.48, 0.72]
+ noise 0 dB	0.38	[0.26, 0.50]
+ blinks 5/min	0.77	[0.65, 0.89]
+ blinks 15/min	0.77	[0.65, 0.89]
+ blinks 30/min	0.77	[0.65, 0.89]
+ noise 0 dB and blinks 30/min	0.39	[0.27, 0.51]

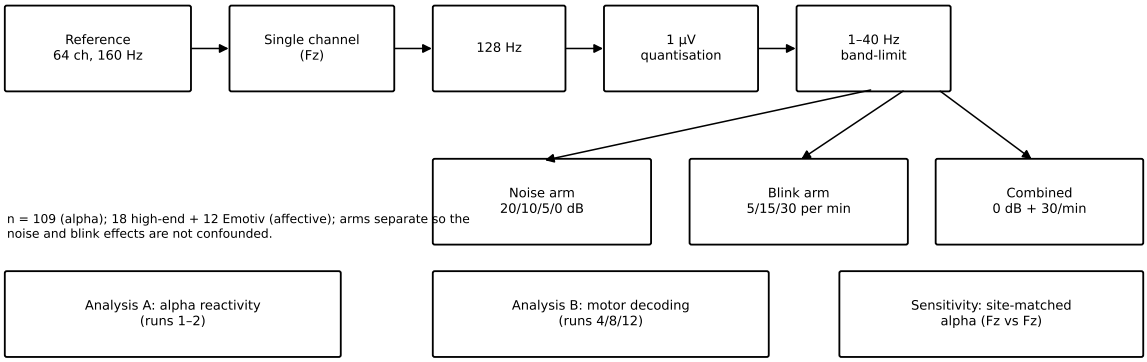


Figure 1: Degradation tree. The reference is the full-montage recording; the single-channel chain is cumulative, while the noise and blink arms branch from the fully degraded chain and are analysed separately.

Noise is the constraint that matters; blinks alone do not shift alpha reactivity

Noise produced a monotonic dose-response: 0.76 at 20 dB, 0.71 at 10 dB, 0.60 at 5 dB and 0.38 at 0 dB. Relative to the clean single-channel chain (the correct baseline for an incremental contamination), the geometric retention was 0.99, 0.94, 0.85 and 0.68; relative to the full-montage reference it was 0.86, 0.82, 0.73 and 0.59. The incremental paired contrasts against the clean chain (which isolate the noise manipulation) were -0.008 ($[-0.011, -0.006]$) at 20 dB, -0.064 ($[-0.075, -0.052]$) at 10 dB, -0.169 ($[-0.197, -0.140]$) at 5 dB and -0.384 ($[-0.441, -0.328]$) at 0 dB; all survive Benjamini–Hochberg correction across the condition family ($q < 0.001$). The three blink arms were almost identical to the clean chain (0.77 at 5 and 15/min, 0.77 at 30/min). The incremental contrast (30 blinks/min minus clean chain) was -0.002 ($[-0.003, -0.001]$; $q = 0.0005$), a change of about 0.2% that is statistically detectable but negligible in size; the combined arm (0.39) was driven by the 0 dB noise it contains. The injected template has almost no energy in 8–12 Hz and the same rate was added to both conditions, so this null is close to guaranteed by construction and should be read as a model check, not as evidence about real blinks; ocular artifacts are known to bias broadband power in ways this additive model does not represent [12]. The template has almost no 8–12 Hz energy and the same rate is added to both conditions, so the null is close to guaranteed and is not evidence about real ocular contamination [11, 7, 37].

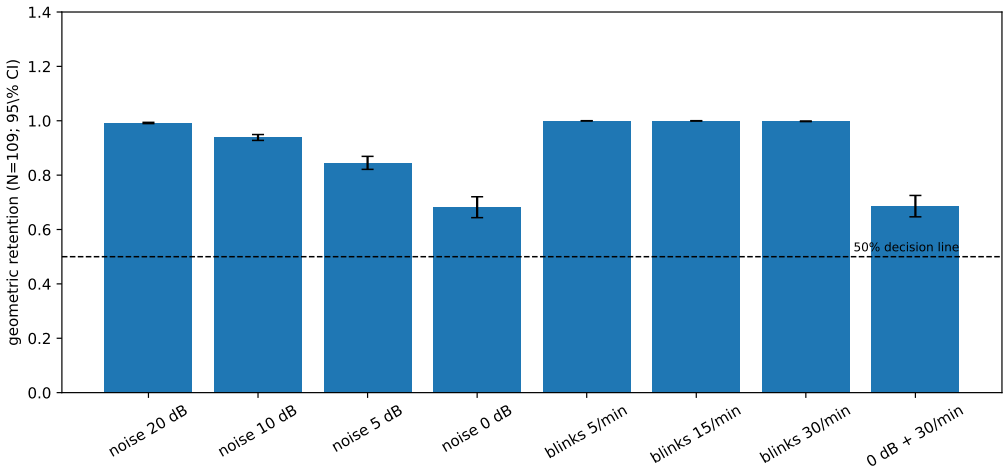


Figure 2: Alpha reactivity retained relative to the clean single-channel chain, by condition ($N = 109$; geometric retention and 95% CI). The dashed line is the planned 50% decision threshold. Noise reduces retention monotonically; the blink arm is a model check and is not a real-blink test.

Affective decoding survives a full montage but not one electrode

Table 2 reports the affective arm on the high-end BrainVision recordings (18 recordings from 20 participants; about 10 trials per quadrant). From the full montage, valence was decoded at 0.69 ($[0.62, 0.76]$), arousal at 0.68 ($[0.60, 0.75]$) and the four quadrants at 0.58 ($[0.49, 0.67]$); the binary intervals exclude chance and the four-class interval is above it. From the same data reduced to the single-channel chain, valence showed no significant evidence of above-chance decoding (0.52; $[0.47, 0.57]$, $p = 0.45$) — arousal to 0.56 ($[0.50, 0.62]$), which remains above chance but far below the montage, and the four quadrants to 0.34 ($[0.27, 0.41]$). The paired full-minus-single differences were $+0.17$ ($[0.10, 0.24]$) for valence, $+0.12$ ($[0.03, 0.20]$) for arousal and $+0.24$ ($[0.14, 0.34]$) for the four quadrants, so the collapse is not just a loss of significance.

A low-cost 14-channel Emotiv headset decoded the same targets at 0.67 ($[0.57, 0.77]$) for valence, 0.71 ($[0.63, 0.79]$) for arousal and 0.56 ($[0.46, 0.67]$) for the four quadrants. On the 12 participants present in both datasets, the low-cost minus high-end difference was -0.07 ($[-0.21, 0.08]$) for valence, $+0.05$ ($[-0.05, 0.15]$) for arousal and -0.04 ($[-0.21, 0.12]$) for the four quadrants. These differences are not statistically distinguishable from zero, but a ± 10-point equivalence test does not pass either (TOST $p = 0.31$, 0.14 and 0.23), so the honest statement is that the low-cost device performs at a similar level in this sample, not that equivalence has been established.

Table 2: Affective decoding by acquisition representation: high-end full montage ($N = 18$ recordings), low-cost 14-channel Emotiv ($N = 12$), and high-end data reduced to one degraded channel ($N = 18$). Balanced accuracy with 95% CI; chance 0.50 (binary) and 0.25 (four-class).

Target	High-end montage	Low-cost 14 ch	Single channel
Valence	0.69 [0.62, 0.76]	0.67 [0.57, 0.77]	0.52 [0.47, 0.57]
Arousal	0.68 [0.60, 0.75]	0.71 [0.63, 0.79]	0.56 [0.50, 0.62]
Four quadrants	0.58 [0.49, 0.67]	0.56 [0.46, 0.67]	0.34 [0.27, 0.41]

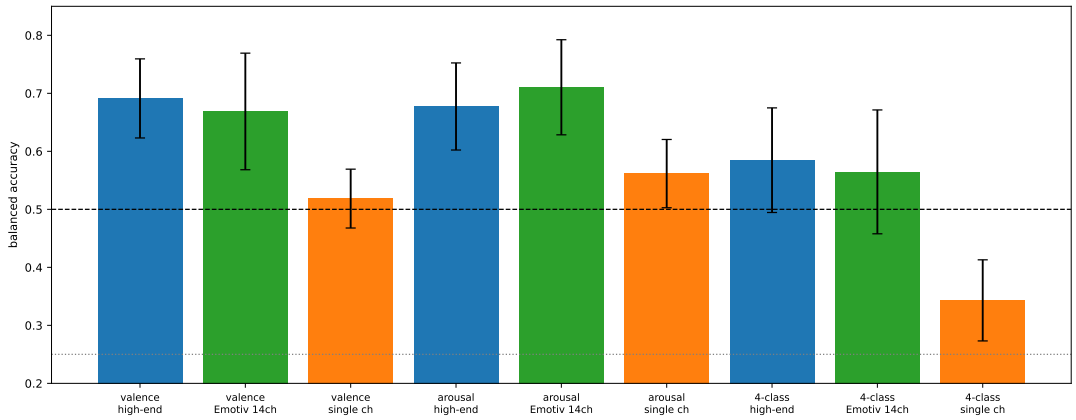


Figure 3: Affective decoding by acquisition representation: high-end montage, low-cost 14-channel Emotiv and single-channel chain. Dashed and dotted lines mark chance for binary (0.50) and four-class (0.25) targets.

Decomposing the affective collapse: representation, site and band

The chain that is harmless for alpha is also harmless for affective decoding (Table 3): resampling to 128 Hz, quantisation and band-limiting left four-class accuracy at 0.58–0.59 and valence at 0.69–0.70. The loss appears when the representation becomes one electrode. Valence was not distinguishable from chance at any of the four sites (0.51–0.53 at Fz, Cz, Pz and Oz); four-class accuracy was 0.33–0.38, above the 0.25 chance level but far below the montage; and arousal was weakly above chance at Fz, Pz and Oz before correction (0.56–0.57; unadjusted $p \approx 0.04$) but not at Cz (0.56, $[0.49, 0.62]$). The single-Fz arousal value survives the six-contrast affective family ($q = 0.0496$) but not the 12-test site sweep ($q \approx 0.079$); we therefore treat the site-level arousal pattern as exploratory. With only four midline sites and three features per site, the bounded claim is that none of these single-electrode representations approached the montage, not that every possible electrode would fail.

A band control changes the interpretation of the full-montage result. With only theta and alpha features the full montage decoded valence at 0.60 ($[0.54, 0.66]$) and the four quadrants at 0.46 ($[0.38, 0.53]$), whereas beta-only features retained 0.70 ($[0.61, 0.79]$) and 0.62 ($[0.53, 0.71]$). A substantial part of the full-montage affective advantage therefore sits in the beta band, where EMG contamination is a known concern [37, 9]. Because beta-only decoding was at least as high as the full montage (0.70 vs 0.69 for valence, 0.62 vs 0.58 for four-class), we do not interpret the montage advantage as purely additive cortical information, and we did not measure EMG directly. This is an interpretation, not a tested mechanism.

Table 3: Decomposition of affective decoding (18 high-end recordings; balanced accuracy). Chance is 0.50 for valence/arousal and 0.25 for four-class.

Representation	Valence	Arousal	Four-class
Full montage, all bands	0.69	0.68	0.58
Full montage, theta+alpha only	0.60	0.63	0.46
Full montage, beta only	0.70	0.70	0.62
128 Hz / quantised / band-limited	0.70	0.68	0.58
Single Fz	0.52	0.56	0.34
Single Cz	0.51	0.56	0.38
Single Pz	0.52	0.56	0.33
Single Oz	0.53	0.57	0.35

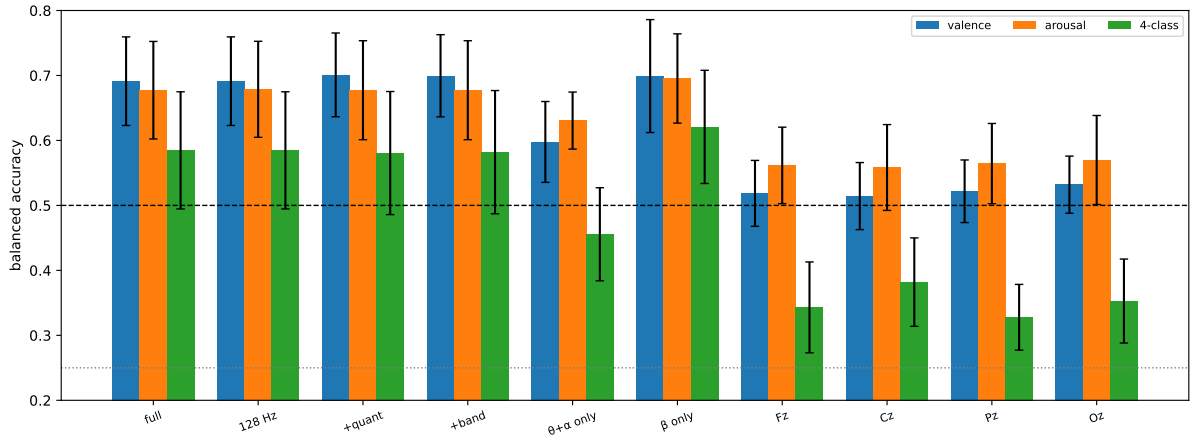


Figure 4: Decomposition of affective decoding by acquisition step and band. The dashed and dotted lines mark chance for binary (0.50) and four-class (0.25) targets.

Reliability depends on the timescale

Short-lag stability of consecutive 2 s alpha windows was $r_{SB} = 0.016$ ($[-0.08, 0.11]$; $N = 109$). This is a short-lag stability measure, not a classical reliability coefficient: odd and even 2 s windows are separated by about two seconds. It says that a momentary alpha estimate is noisy and is consistent with zero. In the ten-subject subset with three repeated imagined-hand runs, block-level alpha power had ICC(2,1)= 0.92, which captures stable between-subject differences. The two are not contradictory, and neither is a test–retest coefficient across sessions; published single-channel test–retest work reports ICCs of roughly 0.76–0.85 for eyes-closed relative power [29], but on a different design.

Exploratory motor decoding was inconclusive

In the ten-subject subset with imagery runs, balanced accuracy for imagined left/right hand was 0.51 ($[0.47, 0.56]$) at Fz and 0.54 ($[0.46, 0.62]$) at the full montage; a CSP reference reached 0.52 ($[0.41, 0.63]$) and a sensorimotor subset 0.55 ($[0.44, 0.66]$). Every interval includes chance, and the full-montage decoder has more features than trials (64 channels, 45 trials). Figure 5 shows the distribution of these accuracies. Because the reference decoder itself did not exceed chance, this arm cannot test the decoding hypothesis; it is reported as **invalid**, not merely inconclusive, and is not used for any claim.

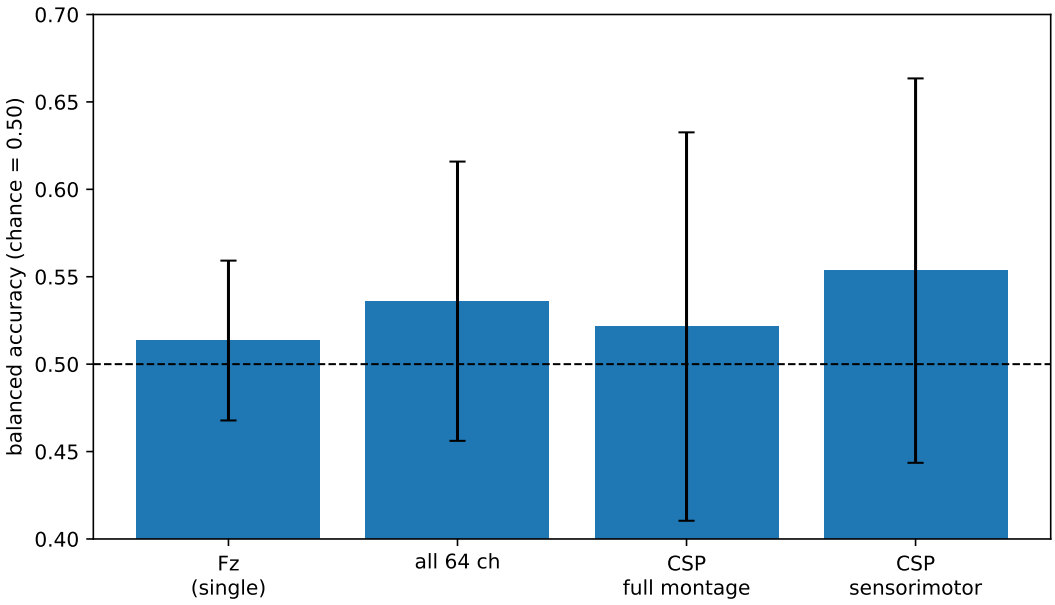


Figure 5: Exploratory motor-imagery decoding ($N = 10$), balanced accuracy by representation. All intervals include chance (0.50).

Decision matrix

Table 4 applies the planned 50% retention rule to the outcomes that were estimated.

Table 4: Claim-level verdicts.

Claim	Evidence	Verdict
Alpha reactivity survives the acquisition chain	Same-site reactivity retention lower bound 0.9996	Supported
Alpha reactivity survives 20–5 dB noise	Clean-chain retention lower bounds 0.99, 0.93, 0.82	Supported
Alpha reactivity survives 0 dB noise	Clean-chain retention 0.68 [0.64, 0.72]	Supported (attenuated)
Combined 0 dB + 30 blinks/min	Clean-chain retention 0.68 [0.65, 0.71]	Supported (attenuated)
Blinks bias alpha reactivity	Clean-chain retention 0.998 in a spectrally orthogonal, symmetric model	Not a real-blink test (H1 untested)
Pre-specified 60% balanced-accuracy floor	Valence full passes (lower 0.62); four-class full fails (lower 0.49); Emotiv valence fails (lower 0.57); arousal full/Emotiv pass	Mixed; reported as auxiliary criterion
Single-trial motor decoding above chance	Reference decoder did not exceed 0.50	Invalid (not testable)
Momentary individual-level alpha reliability	Short-lag stability $r_{SB} = 0.016$	Not supported
Stable individual alpha level	ICC across runs 0.92 (no CI)	Supported (block-level)

Status. The alpha arm uses the full 109-subject dataset. The affective arm uses 18 high-end recordings and 12 low-cost Emotiv recordings; the Emotiv alignment relies on wall-clock timestamps and 8 participants are dropped. The full-montage affective advantage is partly beta-band and may include a muscular contribution. The motor arm is invalid (reference at chance) and underpowered; H5 is untested. These scope limits are stated in the Discussion and are not treated as results.

5 Discussion

The main result is conditional, not a simple asymmetry

The benchmark produced two results that are related but not symmetric in the way a single headline suggests. Alpha reactivity survived the acquisition chain almost intact once the reference was taken at the same electrode (99.99% retained), while the electrode-site decision itself costs about 42% of eyes-closed alpha power. Affective decoding was above chance from the full montage, but it did not survive reduction to one electrode: the valence effect was not detected, arousal remained weak, and four-class accuracy fell to 0.34 (above chance but 0.24 below the montage). The two arms also differ in reference effect size, trial count, feature count and estimator, and the alpha ratio is largely invariant by construction to resampling and band-limiting. We therefore do not present the contrast as a mechanistic law about "broad alpha" versus "distributed affect", and the abstract and conclusion state separate conditional verdicts rather than a single asymmetry. What the benchmark does provide is a shared rule for deciding which claims survive.

What the affective decomposition shows

Three findings constrain the interpretation. First, the acquisition chain that matters for consumer hardware — 128 Hz, 1 µV, 1–40 Hz — did not hurt affective decoding at all; the loss appeared only with one electrode. Second, no single site (Fz, Cz, Pz, Oz) recovered valence or four-class decoding, so the result is not a frontality artefact; arousal, by contrast, stayed weakly above chance everywhere. Third, a band control showed that the full-montage advantage is partly carried by beta-band features: theta+alpha-only decoding fell to 0.60 for valence and 0.46 for the four classes, whereas beta-only decoding retained 0.70 and 0.62. Arousal is the exception: theta+alpha-only still reached 0.63 (93% of the full-montage value), so the beta/EMG caveat applies mainly to valence and the four-class target, and it remains an interpretation rather than a tested mechanism. Beta is exactly where EMG contamination lives [37, 9], so the full-montage number may include a muscular contribution and should not be read as proof that distributed cortical activity encodes affect. We did not measure EMG or test a paired beta-versus-theta/alpha difference, and beta-only decoding was at least as high as the full montage; the claim is therefore an interpretation and a limitation, not a resolved mechanism.

The low-cost Emotiv arm performed at a similar level to the high-end montage in this sample (valence 0.67, arousal 0.71, four-class 0.56; matched $N = 12$ high-end 0.73, 0.66, 0.61) and the within-subject low-minus-high differences were close to zero. A ±10-point equivalence test does not pass, however, so the correct claim is "not statistically distinguishable in this sample", not "equivalent". The comparison also confounds channel count with feature count, recording units, epoch window and condition arm, and 8 of 20 participants are dropped; it is therefore a sanity check, not a device validation. The pre-specified 60% floor is met by Emotiv arousal but not by Emotiv valence, while the high-end arm meets it for valence and arousal but not for the four-class target.

The site-matching lesson

The naive comparison — one electrode against a posterior-dominated scalp average — suggested that a single channel retained about 87–90% of the reactivity effect (geometric and mean-ratio estimators respectively). The same-site analysis showed that this apparent loss was almost entirely a property of the reference: the chain itself cost 0.01%, while choosing Fz instead of a posterior average cost about 42% of eyes-closed alpha power (arithmetic ratio 0.58; geometric ratio 0.49). The two units answer different questions and we report both. This is a known property of scalp power — reference choice and electrode site change absolute band power and can change lateralised contrasts [13] — but benchmark papers routinely ignore it. Our contribution is to quantify its size for single-channel retention and to make same-site referencing the primary estimator. We recommend it as the default for single-channel benchmarks and consumer-EEG validation studies. A related site caveat points in the same direction: Fz is a research midline proxy, not the site used by most single-electrode consumer devices, which record frontally (Fp1 in NeuroSky/MindWave, AF7/AF8 in Muse; [18, 5]). Frontal sites have less posterior alpha and more ocular/muscle contamination, so the measured 42% site cost is likely an underestimate for those devices, and the single-channel affective failure is established here for Fz/Cz/Pz/Oz rather than for every possible electrode. The consumer-EEG reliability comparison is at Fp1 [29], not at Fz, which is why we treat the two estimates as related but not interchangeable.

Noise, bandwidth and the blink null

The noise arm declined monotonically, and every paired contrast survived FDR control; at 0 dB the effect retained about half of its reference size but remained detectable. The practical implication is that recording quality and noise control matter more than a headset's nominal sampling rate or resolution, which changed 8–12 Hz power by less than 1%. The blink result is a model check rather than a null result about real blinks: the injected template has essentially no 8–12 Hz

energy and the same rate was added to both conditions, so a small effect was almost guaranteed. The pre-specified blink prediction (H1) is therefore untested rather than supported. Real blinks are large, frontal and asymmetric between eyes-open and eyes-closed recordings, and that interaction is the next experiment (see Limitations).

Reliability depends on the timescale

Odd and even 2 s windows are separated by about two seconds, so the coefficient is a short-lag stability measure, not a classical reliability coefficient. Its value ($r_{SB} = 0.016$) says that a momentary alpha estimate is noisy; the block-level ICC (0.92) says that stable individual differences are measurable. Published consumer-EEG test–retest work reports ICCs around 0.76–0.85 for eyes-closed relative power [29]; our estimate is not directly comparable because it uses a different design, but the two are not in tension. A consumer product that reports "your alpha right now" is making the first kind of claim and should be judged accordingly.

Positioning against prior work

Our alpha arm overlaps with device-comparison studies, including one that compared four systems on five participants across two visits with Fp1 as the common electrode and explicit eyes-closed/eyes-open and test–retest analyses [28]. We therefore do not claim that nobody has examined consumer spectral reliability. The contribution is the controlled degradation grid, the same-site retention metric, the claim-level decision matrix, and the decomposition of the affective collapse. Work on channel subsets for emotion decoding has shown that a well-chosen small montage can perform surprisingly well [38]; our result adds the complementary finding that a random single site does not, and that beta-band features can inflate full-montage performance.

What the paper can and cannot claim

The paper can claim that, in the datasets analysed here, the acquisition chain that consumer hardware changes is not what limits alpha reactivity, that electrode choice is the dominant cost for single-channel alpha, and that affective decoding from one electrode does not work for valence or quadrant while arousal survives weakly. It cannot claim that any real single-channel consumer device preserves the effect, because no such device was measured; it cannot claim that blinks are harmless, because the blink model is simplified; it cannot claim equivalence between the Emotiv and the high-end system, because the equivalence test did not pass; it cannot claim that the full-montage affective result is purely cortical, because the beta control points to a possible EMG contribution; and it cannot claim that single-channel motor decoding is impossible, because that arm is underpowered. The framing is a methods and reproducibility contribution with conditional findings, not a device validation.

Limitations

The alpha arm uses all 109 subjects. The affective arm uses 18 high-end recordings and 12 low-cost Emotiv recordings; the Emotiv alignment relies on wall-clock timestamps, eight participants are dropped (three empty files, two header-only files and three alignment failures) and the signal is in device units rather than calibrated microvolts. Artifact handling is minimal (no ICA/Autoreject), so beta features may be contaminated by EMG. The blink model lacks topography, is symmetric between conditions, and cannot test the real ocular interaction; H1 is untested. Labels are normative IAPS quadrant assignments, not participants' own ratings, and within-subject cross-validation over about 10 trials per quadrant confounds emotional category with the low-level visual properties of the picture sets; a picture-level (leave-one-picture-out) generalisation test and a label-permutation null were not run. The reference electrode is as provided by the archive; run order is fixed (eyes open before eyes closed); no test–retest ICC across

sessions is available, and the block ICC has no confidence interval. Feature counts differ between the full montage (198) and the single channel (3), and only four midline sites were swept, so the representation comparison confounds information, dimensionality and site; the single-site failure is bounded to those sites. The 0 dB noise level is defined on broadband variance while the outcome is 8–12 Hz, so the in-band SNR is unreported and part of the attenuation is estimator compression; eight of 109 subjects had negative reference reactivity. The analysis plan was not registered, several pre-specified items (Fp1/Cz sensitivity, the 60% practical floor, AUC and median-split outcomes) were dropped or changed, and the study should be read as exploratory in the strict sense. The motor arm is invalid because its reference decoder did not exceed chance (and is also underpowered); the equivalence test is a small-sample TOST whose margin exceeds the achievable half-width for valence and four-class; and the repository has no public DOI or remote yet (the availability statement says it will be deposited). A full-subject run, a real single-channel device, a validated artifact model and a registration are required before the decision matrix is treated as a standard.

Implications for reporting and applied work

For authors, the practical rule is to state the acquisition constraints, compare against a same-site reference, report which band carries the effect, and use the decision matrix to decide whether a claim is supportable. For reviewers, the rule is to ask for that mapping instead of a general judgement about device quality. For applied work, the preliminary result supports two narrow statements before a study is designed: single-channel alpha reactivity is limited mainly by electrode choice and noise, so both drive the protocol; and emotion decoding should not be planned on a single electrode without a pilot, while a low-cost multi-channel headset may be adequate for arousal.

Ethics and positioning

The paper makes no therapeutic, diagnostic or safety claim and evaluates no commercial product's marketing. Its value is a quantified, partly null result, an honest decomposition, and a reporting discipline for single-channel work, released with code and derived data.

6 Conclusion

We presented a reproducible degradation benchmark for single-channel and low-cost EEG and applied it to two open datasets. The results are conditional and, we think, useful. Alpha reactivity survived the acquisition chain almost intact once the reference was taken at the same electrode (99.99% retained; a ratio-of-ratios outcome, so gain-like steps cancel by construction), whereas the electrode-site decision cost about 42% of eyes-closed alpha power and sensor noise reduced the effect monotonically, to about 0.59 of the reference at 0 dB. Affective valence was decodable from a full high-end montage (0.69) but was not detected from a single electrode (0.52); arousal survived weakly (0.56 above chance but far below 0.68), and four-class accuracy fell to 0.34 (above chance but far below the montage). A low-cost 14-channel headset performed at a similar level to the high-end montage without passing a formal equivalence test. Decomposition showed that the acquisition chain was harmless, that valence was not distinguishable from chance at any single site and the four-class result stayed far below the montage, and that beta-band features carried part of the full-montage advantage — a result that cautions against reading the montage number as purely cortical.

The deliverable is a decision matrix, a single-channel reporting checklist, and an open, seeded analysis pipeline that turns public data into the reported numbers. The next version requires the full subject sample, a real single-channel device, a validated artifact model, registration of the analysis plan, and a larger equivalence study. The value of the exercise is that it replaces a

plausible story ("single channels lose information") with a measured one: what a single electrode can and cannot show depends on the analysis, the site and the band, and that boundary can be mapped.

Data and code availability

All datasets are public. The EEG Motor Movement/Imagery Dataset is available from PhysioNet (doi:10.13026/C28G6P). The affective dataset is available from Zenodo under CC BY 4.0 (doi:10.5281/zenodo.14787743). DEAP was not used: its download page was unreachable during this work, and the DREAMER record is access-restricted [16]. The analysis repository (degradation pipeline, analysis scripts, seeds, environment file and the JSON files behind every number) is public at github.com/curiousbrutus/EEG-Benchmark and mirrored on the OSF project page osf.io/8wxqj. A Zenodo archive DOI will be assigned on acceptance; until then the GitHub repository and the OSF project page are the resolvable sources. The commit hash used for this draft is recorded in `results/data_manifest.json`. The single-channel reporting checklist is included as `CHECKLIST.md`. No new human participants were recorded.

Author contributions

Conceptualization, methodology, software, analysis, writing: E.G.

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